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RESEARCH

USH: an efficient real-time distracted driving detection model

He Wang[1] · **Yuan Li**[1]

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Abstract

With the continuous development of intelligent transportation systems, driving safety has become a core societal concern. Distracted driving, one of the leading factors of traffic accidents, has been directly responsible for over 60% of such incidents. To effectively reduce these accidents, real-time monitoring and early warning of distracted driving have become crucial tasks in enhancing traffic safety. In this paper, we proposes a low-latency, high-efficiency network called USH, aiming to quickly and accurately detect driver distraction behavior on low-computation devices. USH employs a stage-wise hybrid modeling approach, introducing convolutional operations and self-attention at different stages of the model to achieve a balance between model accuracy and efficiency. By injecting local information modeling layers into a Feed Forward Network (FFN), the network complexity is significantly reduced while maintaining high detection accuracy. Additionally, USH introduces a single-head self-attention, effectively avoiding the head redundancy of multi-head self-attention, making the model more lightweight and efficient, meeting the detection needs in resource-constrained environments. Experimental results demonstrate that USH achieves a top-1 accuracy of 98.5% on the StateFarm dateset, with an inference latency of only 4.7 ms. Compared to the original model, the proposed USH achieves a 21.6% improvement in computational efficiency, a 146% reduction in model size. These results indicate that USH not only demonstrates excellent performance and stability, but also exhibits significant potential in the field of distracted driving detection.

Keywords Deep learning · Image classification · Distracted driving · Lightweighting

1 Introduction

Statistics from the World Health Organization (WHO) show that approximately 1.19 million people die each year from road traffic accidents, equivalent to one fatality every 60 s [1]. According to relevant research data from Indiana University, more than 80% of traffic accidents are directly or indirectly linked to distracted driving [2]. In the United States, over 3000 deaths occur each year due to distracted driving, and more than 400,000 people are injured as a result of distracted driving [3]. In China, there are 2 million traffic accidents each year, and about 60% of these accidents can be attributed to driver distraction [4]. Distracted driving has been recognized as one of the leading causes of global traffic accidents. It severely impacts road safety and significantly increases both the frequency and severity of accidents.

Any activity or event that diverts the driver's attention away from the primary task of driving is distracted driving. This phenomenon significantly impairing the driver's ability to effectively process environmental information and promptly respond to potential hazards. Based on its nature, distraction can be categorized into three main types: visual distraction (eyes off the road), manual distraction (hands off the steering wheel) and cognitive distraction (attention diverted from the driving). Although each type of distraction independently affects driving performance, in real-world driving, drivers are often affected by multiple types of distractions simultaneously. The combined effects tend to significantly delay reaction time and greatly increase the risk of accidents [5]. For example, a driver may be looking at their phone (visual distraction) while operating the phone (manual distraction) and replying to messages (cognitive distraction). Choudhary shows that conversation and texting increase the accident probability by 3 times and 4 times respectively [6]. Li found that distractions significantly

✉ He Wang
wanghe@sxu.edu.cn

Yuan Li
202322615004@email.sxu.edu.cn

[1] School of Physics and Electronic Engineering, Shanxi University, Taiyuan 030006, Shanxi, China

increase drivers' reaction times, with visual distractions and manual distractions increasing reaction times for rear-end collision events by 0.47 s and 0.38 s respectively [7]. Similarly, A study by Zhang et al. shows that within 4–5 s after viewing information, the driver's operating sensitivity had not yet returned to normal [8]. This indicates that cognitive distraction continues to interfere with the driver. Therefore, reducing distracting behaviors is of paramount importance for reducing accident rates, minimizing economic losses and ensuring the safety of human lives.

Research has shown that improving driver reaction time can effectively reduce traffic accidents by approximately 60% [9]. Therefore, quickly and accurately detecting distracted driving is crucial for driving safety. In recent years, the emergence of deep learning has introduced new directions for this technology and quickly become a research hotspot [10]. Figure 1 presents the general architecture of distracted driving detection system, which is based on computer vision techniques. On this basis, this study specifically focuses on the distraction detection module, highlighted in red in Fig. 1. The other stages are merely described as the background of the overall process and have not been specifically implemented in this study.

However, in practical applications, traditional deep learning models often have complex network architectures and massive parameter volumes, while the computing power and storage capacity of vehicle systems are limited [11]. Thus, how to deploy the model on low-computation devices and improve the operational efficiency has become an urgent issue. To address the above-mentioned issues, we need to consider three important factors: parameters, latency and accuracy. First, we adopt a stage-wise hybrid modeling approach to form a robust design paradigm. Second, in the earlier high-resolution stages, we simplify the model by injecting locality in a unified FFN. Finally, to further enhance the capability of global modeling, we employ efficient self-attention in the later low-resolution stage to capture long-distance dependencies. We named the model USH, which achieves a good balance between accuracy and latency on low-computation devices. Compared to other lightweight networks, USH demonstrates higher accuracy and faster inference speed with a smaller model. Our contributions can be concluded as follows:

1. The model is divided into four stages, using Meta4D and Meta3D respectively, optimizing computational efficiency while retaining global modeling capabilities.
2. Meta4D captures local information by placing additional depth-separable convolutions in the FFN to improve performance with a smaller overhead.
3. Meta3D introduces a single-head self-attention mechanism (SHSA), achieving a good balance between accuracy and latency.

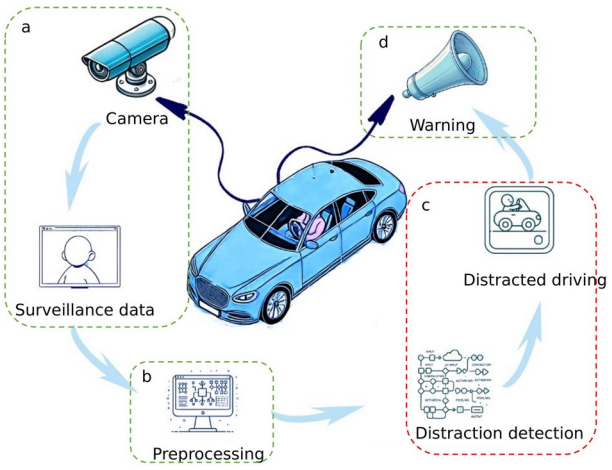


Fig. 1 The general architecture of a distracted driving detection system using computer vision. **a** data acquisition from in-vehicle cameras, **b** data preprocessing, **c** distracted driving detection network, **d** distraction warning for driver

2 Related work

2.1 Distracted driving detection

The classification of data types can be broadly divided into three main categories: physiological data [12], car control data [13], and computer vision data [14]. However, the first two methods have limitations [15]. The first method requires special sensors or devices, which are generally not suitable for prolonged wear and are relatively expensive. The second method is highly influenced by road conditions, vehicle status and the driver's behavior habits. The third method has strong environmental adaptability and good scalability, which can be applied to various scenarios and devices. Applying deep learning methods to driver monitoring can real-time identify and warn against distracted behavior, significantly reducing road traffic accidents.

In recent years, many researchers have studied on distracted driving and proposed their detection methods. Alotaibia et al. proposed a method using the driver's key body points to help Convolutional neural networks (CNNs) identify whether the driver is distracted. The method achieved an accuracy of 94.9% on the StateFarm dataset [16]. Hossai et al. detected distracted driving through facial and hand positioning, achieving an accuracy of 99% [9]. Moreover, some studies focus on reducing the complexity of the network. AlShalfan and Zakariah based on the VGG16 model, utilizing dropout and batch normalization to reduce parameters by 75% [17]. This algorithm achieved an accuracy of over 96%. Wang improved the algorithm by introducing a coordinate attention mechanism, changing the activation

function, and creating a new classifier, achieving an accuracy of 98.77% with a parameter count of only 5.25M [18].

However, most of the existing methods focus on accuracy and network complexity, ignoring latency. Optimizing inference latency is key to enhancing system real-time performance. Wu reduced the computational overhead of the model by pruning the dynamic routing algorithm, achieving an accuracy of 96.5% with a latency of 1.16 s [19]. Tang et al. proposed a lightweight network that introduced a new module containing self-attention, with an accuracy of 99.91% and a FPS of 12.4 on the StateFarm dataset [20]. In addition, Luo et al. proposed a hierarchical bilinear CNN to implement distracted driving recognition, attaining over 90% classification accuracy at 4.21 ms inference latency [21].

These methods, although focused on delay, did not take into account the computing power of the vehicle's system during implementation, nor did they sufficiently balance the relationship between accuracy and delay. Therefore, we aim to optimize the real-time performance of the system low-computation devices while ensuring the accuracy of distraction recognition.

2.2 Lightweight models

In recent years, CNNs have become the primary method for computer vision tasks [22–25]. However, CNNs rely on the local sliding window mechanism, which hinder modeling flexibility. At the meantime, standard CNNs involve a large number of parameters and complex computations, which typically require high computational power and memory. To overcome this challenge, methods such as inverted residual bottlenecks [26], separable convolutions [27] and structural re-parameterization [28] have been proposed to make CNNs more lightweight.

Subsequently, the Vision Transformer (ViTs) achieves state-of-the-art performance on large-scale image recognition tasks [29]. However, ViTs require substantial computation and memory. Therefore, making ViTs more lightweight has become a new goal for researchers. Currently, the mainstream methods can be divided into two categories: one is to reduce computational costs through efficient self-attention [30–34]. The other is to combine ViT with lightweight CNNs to build efficient architectures [35, 36] and accelerate inference speed. For example, Swin Transformer [37] reduces computational costs through sparse attention [38, 39]. MobileViT [40] adopts a hybrid architecture, combining lightweight MobileNet blocks [41] with multi-head self-attention blocks, to achi-eve lightweight design.

CNNs excel at extracting local features, with strong spatial invariance and computational efficiency [42]. However, the limitation of CNN is that it can only capture local information and is difficult to model long-range dependencies. ViTs can effectively model global dependencies through the self-attention mechanism, but its computational overhead is large and its ability to extract local features is weak [43]. Based on this, we adopt a CNN-Transformer hybrid structure, aiming to simplify the model structure and reducing the computational complexity.

3 Method

This section elaborates the proposed approach. We start by briefly summarizing the structure of USH in Sect. 3.1. In Sect. 3.2, early convolutions are introduced instead of patch embedding to reduce computational costs. In Sect. 3.3, we reduce the network complexity by injecting local information modeling layers in the FFN. Finally, in Sect. 3.4, we introduce the SHSA, which effectively reduces the redundancy of the multi-head self-attention.

3.1 Overview

USH adopts the hierarchical pyramid structure to handle feature information at different scales and reduce computational complexity [44–48]. At the lower levels, the model focuses on fine-grained details and local features. At the higher levels, the model better integrates global information, improving the model's robustness and generalization capability [49]. The overview of USH is shown in Fig. 2a.

Memory access overhead is a critical factor affecting model speed [50–53]. As shown in Fig. 3, many operations in transformers are memory-inefficient, especially the frequent reshaping and element-wise addition in MHSA [54]. The memory-bound operations are denoted by the red regions. Therefore, USH uses a CNN-Transformer structure. In the early stages, a unified FFN is used for efficient computation, avoiding introducing computationally expensive self-attention too early, and ensuring lower latency. In the later stages, the self-attention is adopted to provide global modeling capabilities for complex tasks [55]. Next, we will delve into the details of the network design.

3.2 Early convolutions for stem

ViTs typically use a patchify operation as the stem, dividing the input image into non-overlapping patches, but this method significantly increases computational cost. USH draws on early convolution methods, using fast down sampling instead of non-overlapping patches [56]. It consists of two hardware-efficient 3×3 convolutions. By stacking these 3×3 convolutions, it is possible to perform efficient feature extraction while reducing the computational load.

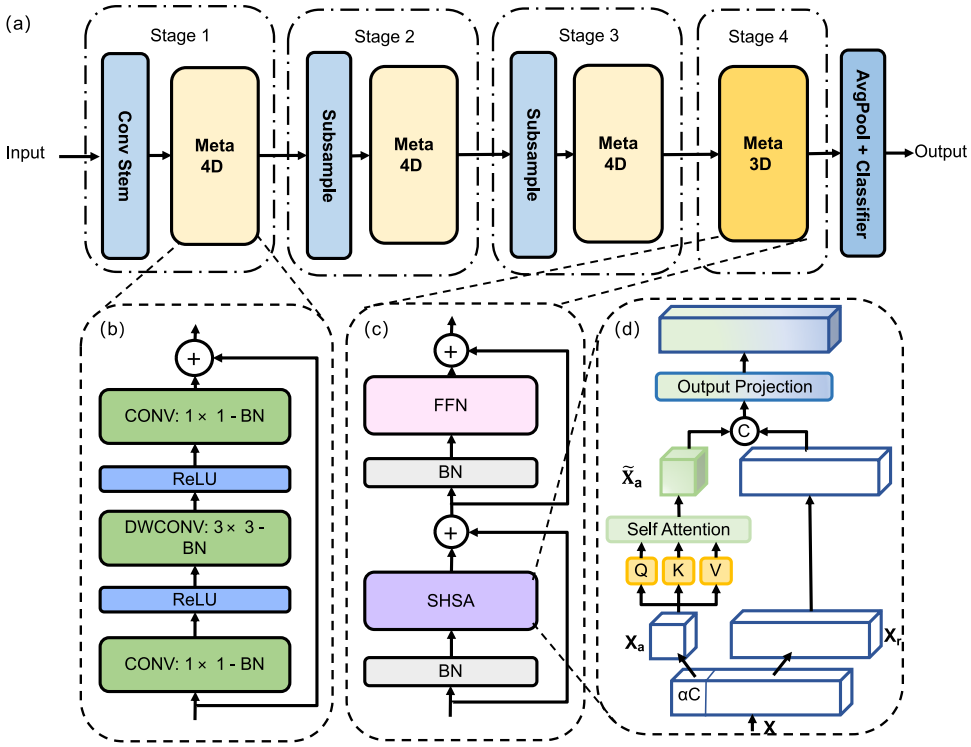


Fig. 2 Overview of USH. **a** Architecture of USH, **b** Meta 4D block, **c** Meta 3D block, **d** single-head attention

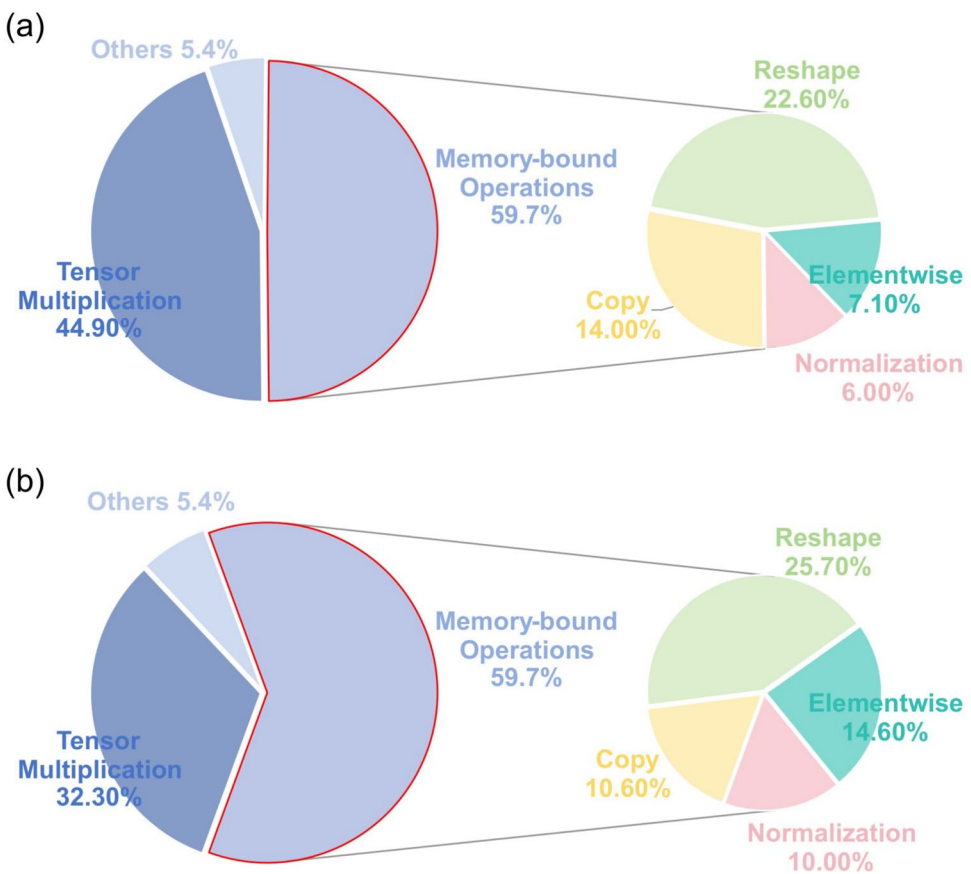


Fig. 3 Runtime profiling on Swin-T and DeiT-T based on [54]. The memory-bound operations are denoted by the red regions. **a** Swin-T, **b** DeiT-T

3.3 The unified feed forward network

Recent research suggests that injecting local information modeling layers into the FFN can improve performance with minimal overhead, enhancing the model's robustness [57, 58]. Drawing on this idea, USH removes the local token mixer (Fig. 4b) in Poolformer [59] and place an additional 3×3 depth-wise convolution in the FFN to capture local information. Finally, a locality-enabled FFN is obtained. We name it U-FFN. We apply U-FFN to the Meta4D of the network.

In Fig. 4, we discuss the differences between Meta 4D and the local token mixer. The local token mixer contains two sub-modules with two skip connections, while our Meta 4D contains only a single residual sub-module. That is more general and hardware friendly.

3.4 Single-head self-attention

Recent studies reveal that there is a high similarity between the heads in the attention map, leading to computational redundancy. Research shows that when using 6 heads, the feature similarity reached 64.8%; whereas, when reduced to 3 heads, the feature similarity decreased to 49.5% [60]. This indicates that using too many heads in MHSA can lead to inefficiencies and increased memory overhead.

Based on the above analysis, USH adopts the SHSA, as shown in Fig. 2c, d. It applies a single-head attention layer to a subset of input channels ($C_p = \alpha C$). Specifically, the input feature map X is divided into X_a and X_r: X_a is used to compute the self-attention and X_r remains unchanged. Finally, the attention-processed features $\tilde{X_a}$ is concatenated with the unprocessed channel features X_r and then projects across all channels. We set α to 1/4 as default. Formally, the SHSA layer can be described as:

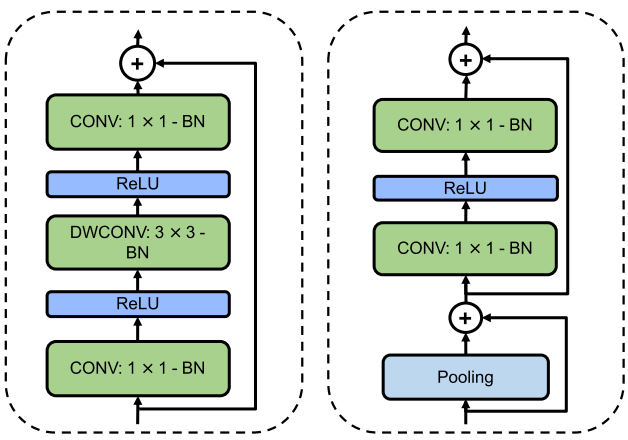


Fig. 4 Comparison between Meta 4D and the local token mixer

$$X_a, X_r = \text{Split}(X, [C_p, C - C_p]) \tag{1}$$

$$\tilde{X_a} = \text{Attn}(X_a W_q, X_a W_k, X_a W_v) \tag{2}$$

$$\text{SHSA}(X) = \text{Concat}(\tilde{X_a}, X_r) W_o \tag{3}$$

where W_q, W_k, W_v and W_o are projection weights, Split$(\cdot)$ is the splitting operation, and Concat $(\cdot)$ is the concatenation operation. We apply SHSA to the Meta3D of the network.

4 Experiments

This section provides a detailed description of the experimental setup, the environment configuration and the experimental results. Through empirical validation and comparison with other relevant studies, the effectiveness and progress of this research are demonstrated.

4.1 Experimental description

Dataset description: The StateFarm Distracted Driving Dataset, a competition dataset on the competition platform Kaggle, is the first publicly downloadable dataset for recognizing distracted driving behavior. The StateFarm dataset contains 22,424 labeled training images. The dataset contains 10 types of driving behaviors from 26 drivers: Safe Driving (c0), Texting—Right (c1), Phone—Right (c2), Texting—Left (c3), Phone—Left (c4), Operating the Radio (c5), Drinking (c6), Reaching behind (c7), Hair & Makeup (c8), and Talking to Passenger (c9).

Evaluation method: The experiment uses K-fold cross-validation. It is used to evaluate the model's performance in classifying data after the training set is randomly shuffled. We set $K = 5$, dividing the dataset into 5 subsets. Each subset is used as the testing set in turn, while the other 4 subsets are used as the training set. Figure 5 is the histogram of the class distribution for the training set.

Evaluation metrics: To ensure an accurate evaluation of the performance of USH, we employ multiple evaluation metrics. These metrics include TOP-1 accuracy, Precision, Recall, and F1 score. In addition, when comparing with other lightweight models, we use latency, Params and FLOPs to evaluate the computational complexity.

Data augmentation: To prevent over-fitting and improve model performance, we apply data augmentation strategies before training, including eight operations such as random flipping, random erasing, edge enhancement, and others. Figure 6 presents a comparative analysis of the image transformations before and after applying eight distinct data augmentation strategies. This can create more diverse data

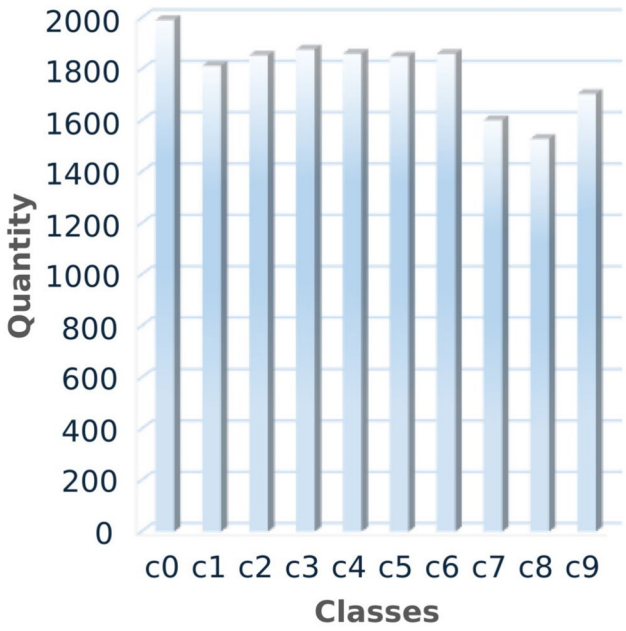


Fig. 5 The histogram of the class distribution

inputs, enhance the model's robustness in practical applications and improve the model's generalization ability.

4.2 Experimental configurations

The experimental setup is shown in Table 1. We use the PyTorch deep learning framework. Train on a single NVIDIA RTX3060Ti GPU with 16 GB of RAM. The initial learning rate is 0.0001. The weight decay is 0.05. The batch size is 32. Use the Adam optimizer. Use a cosine learning rate scheduler with 3 linear warm-up steps.

4.3 Experimental results

4.3.1 The experimental results of USH

Table 2 presents the classification results of the fivefold cross-validation on the testing set. USH achieves an average accuracy of 98.5% on the annotated testing set. At the same time, the TOP-1 accuracy for each fold exceeds 98%, and the TOP-5 accuracy exceeds 99%. The results indicate

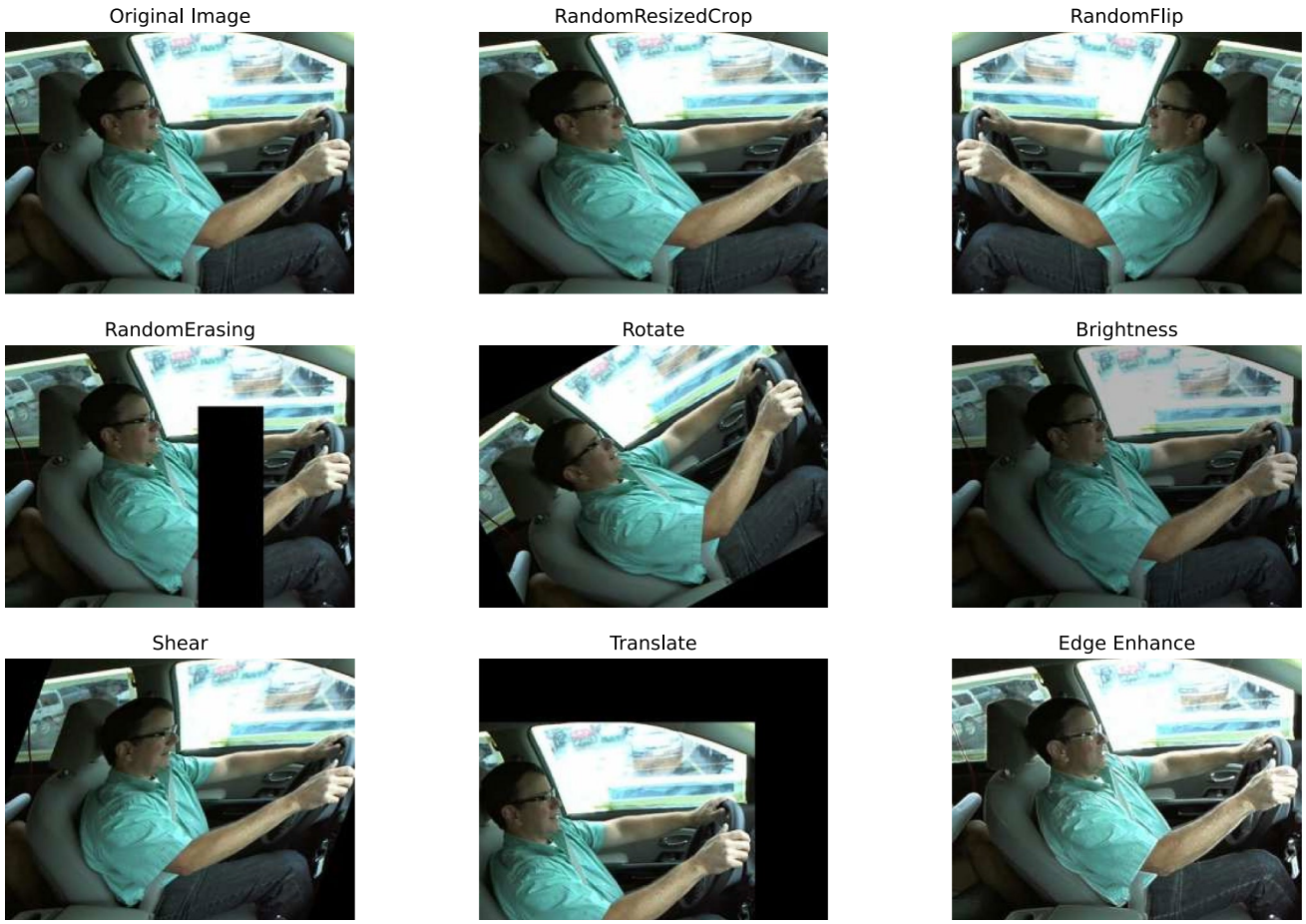


Fig. 6 Comparison of changes before and after the augmentation

Table 1 Experimental configurations

CPU	Intel(R) Core(TM) i7-11700 @ 2.50 GHz
GPU	NVIDIA RTX3060Ti
RAM	16G
Framework	PyTorch
Language	Python 3.6
Learning rate	0.0001
Batch size	32
Weight decay	0.05
Optimizer	Adam

that USH performs exceptionally well in the current task, demonstrating strong generalization ability and stability.

Figure 7 shows the classification performance of each fold. Obviously, the accuracy for 10 driving behavior classes in all fold reaches over 97%. Among them, Fold-5 achieves the highest accuracy for class c0, reaching 99.2%. In the fivefold cross-validation, USH shows small fluctuations in accuracy on c2, c3, c4 and c5, indicating strong stability and high generalization ability for these classes. In contrast, there are larger fluctuations on c7. It is possible that this class has a larger number of complex samples in certain subsets of data.

4.3.2 Comparison results with other models

In addition, we compare USH with several state-of-the-art lightweight models. Table 3 presents the comparison results. USH outperforms most models in TOP-1 accuracy, only slightly trailing behind MambaOut-Femto and MILA-T.

Table 4 presents the comparison results in terms of parameters and latency. USH achieves the best performance in terms of latency and FLOPs, and also has significant advantages in terms of parameters. USH's inference speed is 18.6% faster than Efficientformer-l1. It is also 3.3 times faster than Swin-T. Although the accuracy of USH is slightly lower than that of MILA-T, it is far better than MILA-T in terms of FLOPs, parameters, and latency. Especially, The latency of USH is 2.1 ms less than that of MILA-T.

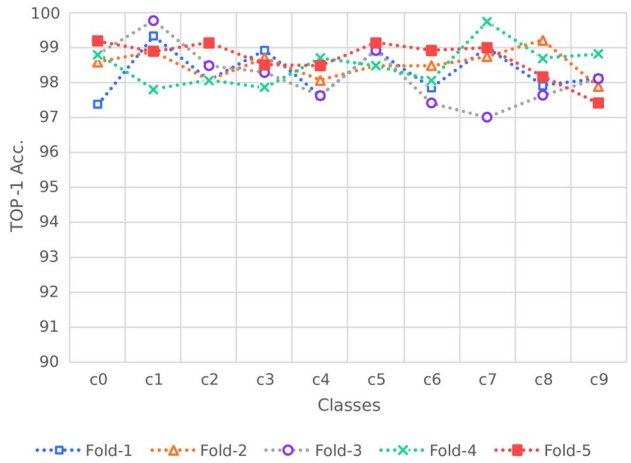


Fig. 7 The performance of each fold

MobileNetV3-S has the optimal number of parameters, but its Top-1 accuracy and latency are both lower than USH.

Figure 8 shows the confusion matrix of each model, demonstrating the classification performance of the models across various classes. The overall performance of the USH is outstanding, achieving a good classification performance in most classes. The performance in c0, c1, c3, and c8 is particularly excellent, with accuracy exceeding 98.5%. In the prediction process, c9 performed relatively poorly, and it has the highest probability of being misclassified as category c0. This indicates that USH may have some difficulty distinguishing between c0 and c9. This may be due to a high similarity in features between these two classes, making it challenging for the model to differentiate effectively during prediction. All models perform excellently in recognizing c0. Among them, MILA-T has the highest accuracy in recognizing category c0, reaching 99.3%.

Table 5 shows the results compared to SOTA methods on the State Farm dataset. MAVIT has the highest accuracy, reaching 99.64%, but has a longer delay. OLCMNet's delay is close to that of USH, but its accuracy is slightly lower. Overall, USH has a higher parameter count, but has achieved the best balance between speed and accuracy.

Table 2 Experimental results of using 5-folds cross validation on StateFarm dataset

Folds	Top-1 Acc. (%)	Top-5 Acc. (%)	Precision (%)	Recall (%)	F1-Score (%)
Fold-1	98.31	99.89	98.34	98.30	98.27
Fold-2	98.52	99.67	98.54	98.50	98.51
Fold-3	98.24	99.42	98.28	98.21	98.24
Fold-4	98.48	99.79	98.53	98.51	98.51
Fold-5	98.71	99.80	98.75	98.69	98.72
Avg	98.49	99.67	98.52	98.48	98.50

Table 3 Comparison results with other models

Model	Top-1 Acc. (%)	Top-5 Acc. (%)	Precision (%)	Recall (%)	F1-Score (%)
MobileNetV3-S [61]	98.40	99.33	94.84	94.79	94.78
Efficientformer-l1 [55]	96.90	99.51	96.93	96.92	96.91
DeiT-T [43]	97.33	99.80	97.35	97.30	97.33
MobileViTv3-XS [62]	97.46	99.79	97.45	97.43	97.43
PoolFormer-S12 [59]	97.76	99.69	97.78	97.74	97.76
SwiftFormer-S [63]	98.39	99.82	98.41	98.39	98.40
MambaOut-Femto [64]	98.75	99.82	98.77	98.74	98.74
MILA-T [65]	99.24	99.96	99.24	99.23	99.24
USH(ours)	98.49	99.67	98.52	98.48	98.50

Table 4 Comparison results in terms of parameters and latency

Model	Params (M)	FLOPs(G)	Top-1 (%)	latency (ms)
MobileNetV3-S	**2.5**	1.7	94.8	5.6
Efficientformer-l1	12.3	2.6	96.9	6.0
DeiT-T	5.9	2.6	97.3	7.2
MobileViTv3-XS	3.0	1.8	97.5	8.5
PoolFormer-S12	12.0	3.9	97.8	10.2
SwiftFormer-S	6.1	2.1	98.4	6.3
MambaOut-Femto	7.0	2.4	98.8	6.2
MILA-T	24.9	4.2	**99.2**	6.8
USH(ours)	5.0	**1.1**	98.5	**4.7**

Bold formatting denotes column-wise optimum values

4.3.3 Ablation study

Ablation study for attention module: We evaluated the impact of several efficient attention mechanisms on module performance, aiming to explore the balance between model accuracy and efficiency through different attention methods. Table 6 presents the effects of various attention on the feature extraction module. On the StateFarm dataset, SHSA far outperforms other attention module in Params and Top-1 accuracy, and also has a slight advantage in latency. CGA has the minimum FLOPs, but its TOP-1 is lower than other methods.

Ablation study for USH: To further demonstrate the effectiveness of the proposed framework, we test the impact of the two modules on model performance. The experimental results are shown in Table 7. After joining U-FFN, the model's parameters has decreased by nearly half, and latency has also been significantly reduced. At the meantime, SHSA slightly improved the model's TOP-1 accuracy. The best result was achieved when using two modules simultaneously. These results demonstrate the ability of the proposed network in capturing both local and global feature representations.

5 Conclusion

In this work, we propose USH, a hybrid architecture integrating Transformer and CNN, designed to recognize distracted driving behaviors. The aim is to maintain a balance between detection accuracy and inference speed on low-computation devices. We introduce U-FFN to reduce network complexity and adopt SHSA to enhance computational efficiency. We extensively benchmark and compare our work with existing studies. The experiments show that USH achieves a Top-1 accuracy of 98.5% on the StateFarm dataset, with an inference speed of only 4.7ms. It is superior to most existing models. This approach strikes a good balance between efficiency and accuracy.

Table 5 The results compared to SOTA methods on the StateFarm dataset

Model	GPU	Params (M)	FLOPs (G)	Top-1 (%)	latency (ms)
OLCMNet [66]	NVIDIA GTX1080Ti	2.8	0.7	89.5	4.7
MDAM [67]	NVIDIA RTX4090	1.4	–	93.7	9.9
ATD2Net [68]	NVIDIA RTX4090	3.2	–	98.2	7.1
RepHBP [21]	NVIDIA RTX3070	0.4	7.1	90.2	7.8
MAViT [69]	NVIDIA RTX4070	**0.2**	**0.5**	**99.6**	14.2
SHSA(ours)	NVIDIA RTX3060Ti	5.0	1.1	98.5	**4.7**

Bold formatting denotes column-wise optimum values



Fig. 8 Result of confusion matrix

Table 6 The performance of different attention methods

Attention	Params (M)	FLOPs (G)	Top-1 (%)	latency (ms)
BRA	5.5	1.1	97.8	5.2
MHA	5.8	1.2	97.6	5.3
CGA	5.7	**1.0**	96.2	4.9
SHSA(ours)	**5.0**	1.1	**98.5**	**4.7**

Bold formatting denotes column-wise optimum values

Table 7 Ablation for our model

U-FFN	SHSA	Params (M)	FLOPs (G)	Top-1 (%)	latency (ms)
×	×	12.3	2.6	96.9	6.0
×	✓	11.4	2.4	98.2	5.7
✓	×	5.8	1.2	97.9	5.1
✓	✓	**5.0**	**1.1**	**98.5**	**4.7**

Bold formatting denotes column-wise optimum values

Despite the strong performance of the USH across all indexes, some limitations remain. For instance, the accuracy are still somewhat lower than other more light-weight models. The model's real-world applicability has yet to be tested. In the future, we intend to implement the model on mobile and embedded devices to monitor distracted information in real-time and confirm its validity. Our ultimate goal is to enhance the model performance by incorporating light-weight network architectures and variants of self-attention to better identify distracted driving features.

Author contributions Y.L. wrote the main manuscript text; Y.L. and H.W. participated in the experimental design and data collection; Y.L. prepared Figures and Tables; H.W. provided critical revisions and edited the manuscript for clarity and style; H.W. provided the final approval of the manuscript before submission.

Data availability The data supporting the findings of this study are available in the State Farm Distracted Driver Detection competition dataset on Kaggle: https://kaggle.com/competitions/state-farm-distracted-driver-detection

Declarations

Conflict of interest The authors declare no competing interests

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